UNIT 2 · TOPIC 8(C)

Combine Harvester Design

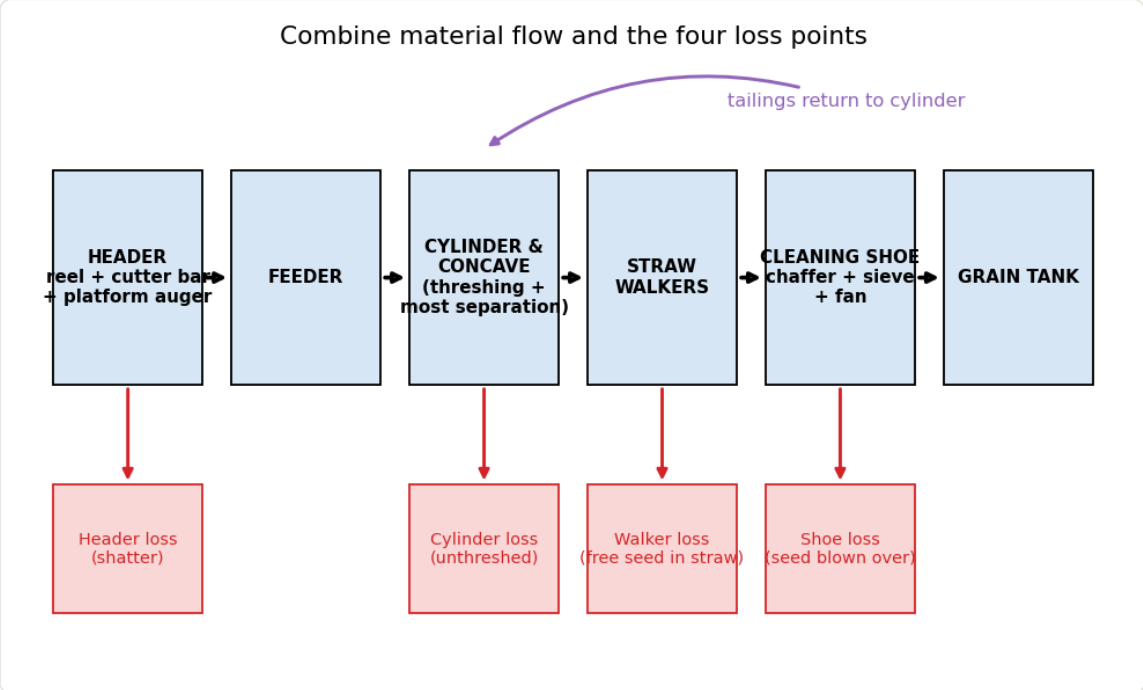
Syllabus topic: Design of harvesters and threshers

Functional elements and material flow

A combine (combined harvester–thresher) cuts/gathers, conveys, threshes, separates, cleans and handles grain. Material flows through:

1. **Header** — reel + cutter bar + platform auger
2. Feeder conveyor
3. Cylinder & concave
4. Beater
5. Straw walkers (rack)
6. Cleaning shoe — chaffer sieve + shoe sieve + fan
7. Clean-grain auger/elevator
8. Grain tank

Partially threshed material returns to the cylinder via the **tailings elevator**.



Combine material flow and the four loss points

Header and reel design

- The reel pushes standing crop onto the cutter bar, holds it while it is cut, then sweeps it onto the platform. Use a **bat reel** for standing crop and a **pickup reel** (spring-tine, cam-controlled bat angle) for lodged/down and tangled crops.

$$\text{Reel index } \lambda = \text{reel peripheral speed} \div \text{forward speed} \approx 1.25\text{–}1.5$$

- λ too high → shattering/header loss; λ too low → crop pushed forward, ragged cutting.

- Set the reel axis ahead of and above the cutter bar, at a height where the bats strike below the heads.
- **Header height:** cut just low enough to gather all heads (less straw into the machine = higher capacity, lower walker/shoe losses).
- **Header (gathering) losses** = shattered free seed + cut/uncut heads dropped — often the **largest single loss** in shattering-prone crops.

Threshing and separating

- The rasp-bar cylinder is standard; change the cylinder speed (variable drive) for the crop, not the diameter.
- Most threshed grain separates at the concave and finger grate. Oscillating **straw walkers** recover the remaining free grain from the straw — multi-section crank-driven racks (**~200 rpm**) toss the straw rearward while grain falls through. Walker area governs capacity in straw-heavy (high MOG) crops.
- The **grain-to-MOG ratio** (grain : Material-Other-than-Grain) characterises crop load; combine capacity is quoted as kg grain/s or kg MOG/s at a specified loss level (ASAE).

Cleaning shoe

- The **chaffer** (upper, adjustable-lip) sieve passes grain + tailings; the **shoe** (lower) sieve gives final cleaning; the fan blast winnows chaff over the rear.
- Air **too high** → free seed blown over (shoe loss). Air **too low** → chaffy grain and tailings overload (recirculation → more damage).
- Tailings (unthreshed heads) are re-threshed at the cylinder.

Losses and adjustment logic

Loss point	What it is	Adjust
Header	Shatter/uncut heads	Reel speed/height, forward speed
Cylinder	Unthreshed seed in straw	Cylinder speed/clearance
Walker/rack	Free seed carried in straw	Feed rate too high, walker overload
Shoe	Seed blown/carried over	Fan and sieve settings

- Measure losses by catching and weighing the discharge over a known area; acceptable total loss is commonly **~2–3%**.
- **Seed damage** occurs almost entirely in the threshing unit (visible + internal/germination damage). Minimise it with the lowest adequate cylinder speed and wider clearance; keep grain-tank/auger speeds moderate.

Worked examples

Ex.4 Reel design. Forward speed 5 km/h (1.39 m/s), reel index 1.3 → reel peripheral speed = 1.81 m/s; reel diameter 1.0 m → $N = 60 \times 1.81/(\pi \times 1.0) \approx$ **34.5 rpm**.

Ex.5 Walker loss. Free seed on the ground behind the walkers over 1 m² = 30 wheat seeds (~40 mg each) = 1.2 g/m² = 12 kg/ha; at 4 t/ha yield → **0.3% walker loss**.

Practice MCQs

Q167. In a combine, most of the threshed grain first separates at the:

(a) Straw walkers
(b) Concave and grate
(c) Chaffer sieve
(d) Grain tank

Q168. Straw walkers recover:

(a) Unthreshed heads only
(b) Free grain remaining in the straw mat
(c) Chaff
(d) Weed seed

Q169. The tailings elevator delivers tailings to the:

(a) Grain tank
(b) Cylinder for re-threshing
(c) Straw spreader
(d) Fan

Q170. The upper adjustable-lip sieve of the cleaning shoe is the:

(a) Concave
(b) Chaffer
(c) Scalper
(d) Grader

Q171. Excessive fan blast in the cleaning shoe causes:

(a) Chaffy grain
(b) Free seed blown over the shoe (shoe loss)
(c) Unthreshed heads
(d) Header loss

Q172. Reel index is the ratio of:

(a) Reel diameter to cutting width
(b) Reel peripheral speed to forward speed
(c) Reel rpm to cylinder rpm
(d) Crop height to reel height

Q173. Recommended reel index for combining standing cereals is about:

(a) 0.5–0.8
(b) 1.0
(c) 1.25–1.5
(d) 2.5–3.0

Q174. Too high a reel speed causes mainly:

(a) Unthreshed loss
(b) Grain shattering at the header
(c) Walker loss
(d) Chaffy sample

Q175. A pickup reel (spring-tine, cam-controlled) is used for:

(a) Tall standing crop
(b) Lodged/down crop
(c) Root crops
(d) Green fodder

Q176. At 5 km/h with reel index 1.3 and 1.0-m reel diameter, reel speed ≈

(a) 15 rpm
(b) 35 rpm
(c) 70 rpm
(d) 140 rpm

Q177. 'MOG' in combine capacity ratings means:

(a) Moisture of grain
(b) Material other than grain
(c) Mass of grain
(d) Margin of grain

Q178. Unthreshed seed discharged with straw from a combine is classified as:

(a) Header loss
(b) Cylinder loss
(c) Shoe loss
(d) Rack loss

Q179. In shattering-prone crops, the largest share of total combine loss is usually the:

(a) Header (gathering) loss
(b) Cylinder loss
(c) Shoe loss
(d) Leakage

Q180. Nearly all seed DAMAGE in a combine occurs at the:

(a) Reel
(b) Threshing (cylinder–concave) unit
(c) Grain tank
(d) Straw spreader

Answers

Q167 — (b) Concave and grate

Q168 — (b) Free grain remaining in the straw mat

Q169 — (b) Cylinder for re-threshing

Q170 — (b) Chaffer

Q171 — (b) Free seed blown over the shoe (shoe loss)

Q172 — (b) Reel peripheral speed to forward speed

Q173 — (c) 1.25–1.5

Q174 — (b) Grain shattering at the header

Q175 — (b) Lodged/down crop

Q176 — (b) 35 rpm — *Reel peripheral speed = 1.3 × (5/3.6) = 1.81 m/s; N = 60v/πD = 60×1.81/3.1416 ≈ 34.5 rpm.*

Q177 — (b) Material other than grain

Q178 — (b) Cylinder loss

Q179 — (a) Header (gathering) loss — *Header/gathering (shatter) loss commonly dominates in crops like sesame, mustard, soybean; it is controlled by reel setting, forward speed and timely harvest.*

Q180 — (b) Threshing (cylinder–concave) unit

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 14 (Hay Harvesting: Mowing) and Ch. 17 (Grain and Seed Harvesting); Jagdishwar Sahay — Elements of Agricultural Engineering; BIS thresher standards (IS 9129 safety); NPTEL Farm Machinery lectures.